

RiVault

**A Secure AI Infrastructure for Scientific Computing
and General-purpose Applications at RIKEN**

Motivation for internally hosted Generative AI

Security Risk Factors

- **Humans** occasionally make mistakes
 - 2023: Samsung workers accidentally leak trade secrets via ChatGPT
- AI software **supply-chain** attacks
 - **Ollama, vLLM, pyTorch, ...** are build on dozens or hundreds of dependencies
- AI **models** can contain malicious code
 - Certain methods of de-/serialization (eg. torch.save, pickle, ...) executes code
- AI **companies** want all our data

Study: 77% of Businesses Have Faced AI Security Breaches

AI systems are particularly vulnerable to security breaches, which is why shoring up your defenses is key in 2024.

Written by
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March 22, 2024

Protect AI's September 2024 Vulnerability Report

Dan McInerney & Marcello Salvati

September 19, 2024 • 3 minute read



Most Recent

Texas Beats New York on Logistics Pay Increases & Work-Life Balance
Adam Rowe • 14 hours ago

Src: tech.co

					Recommendations
	h2o-3				Upgrade to version 3.46.0.1
CVE-2024-5386	Account Hijacking via Password Reset Token Leak	Critical	9.6	Yes	Upgrade to version 1.2.14
CVE-2023-6038	LFI in h2o-3 API in h2o-3	Critical	9.3	Yes	Upgrade to version 3.46.0.1
CVE-2024-5328	SSRF through backend endpoint auth api in lunary	High	8.6	Yes	Upgrade to version 1.2.14

Src: protectai.com

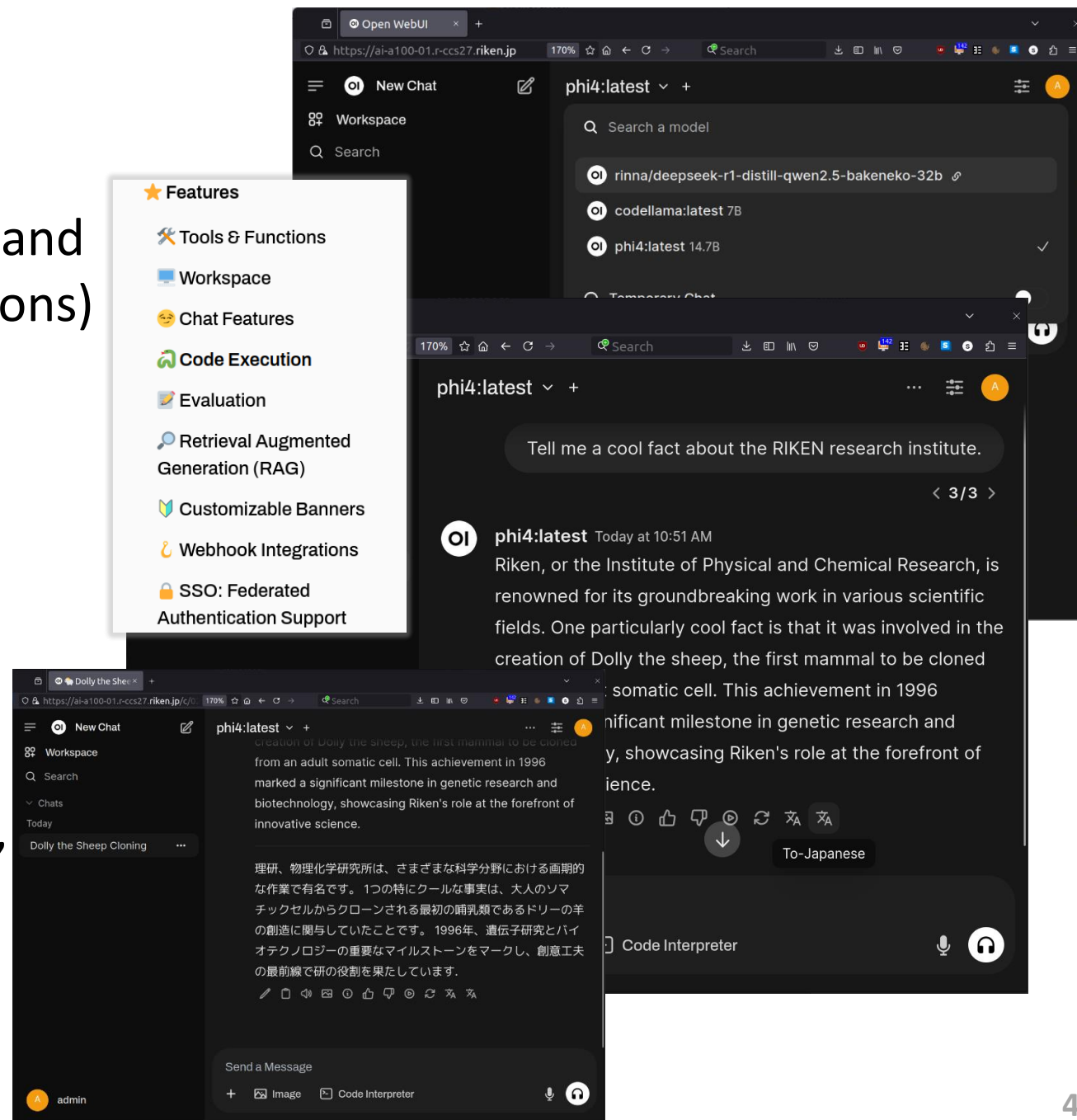


Src: blackfog.com

WebUI

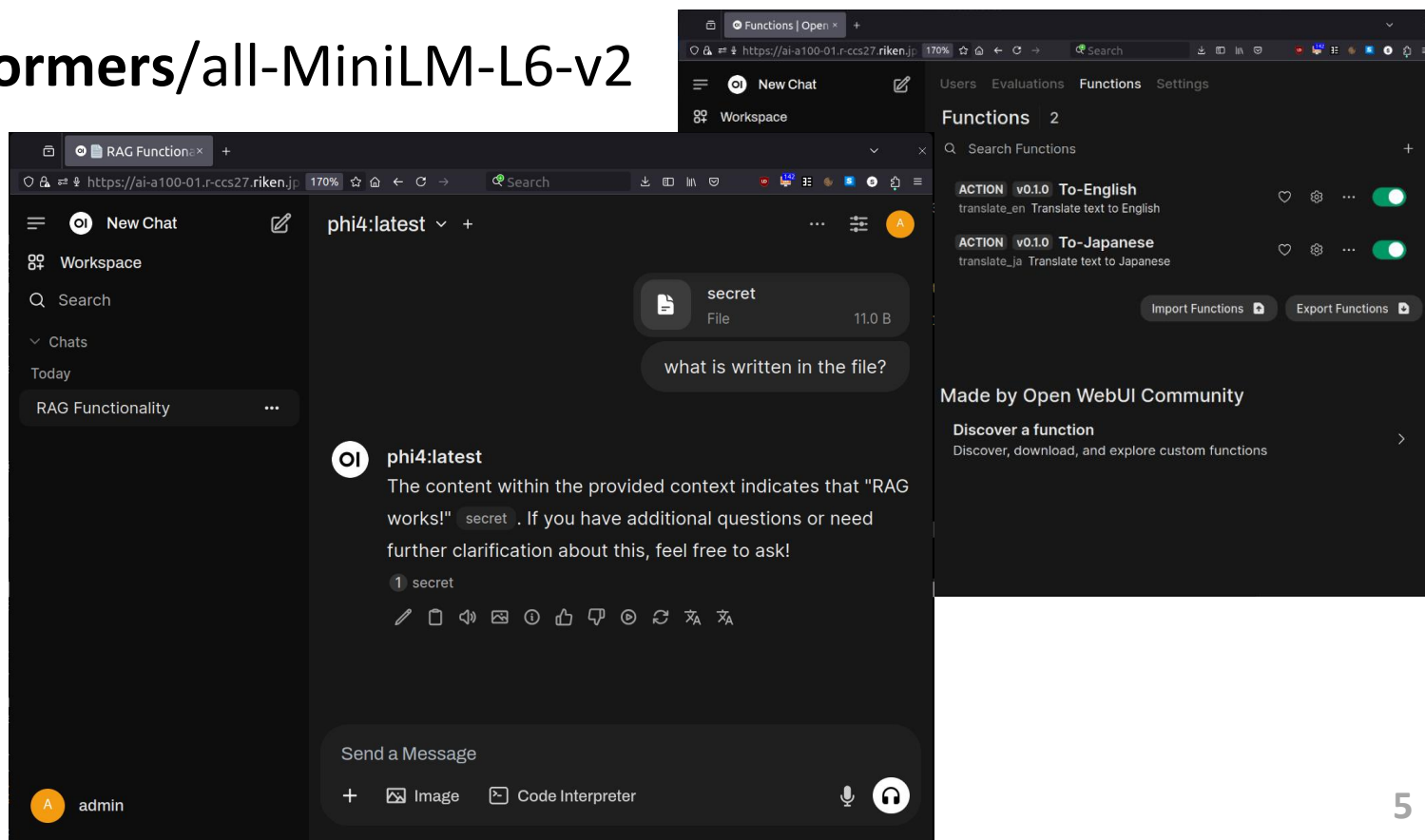
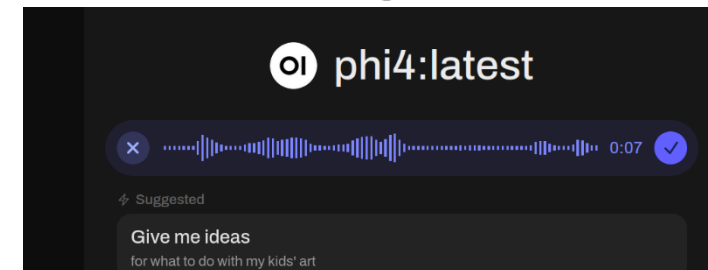
User-facing Web UI

- Self-hosted models
- Zero data leakage: internal, secure, and no internet access (with few exceptions)
- Very similar design to **OpenAI interface**
- Supports **various features** and **multiple chats/contexts**
- Easy to select from available **models via dropdown**
- Customizable via scripts/"functions" (e.g. our en<->ja translations)



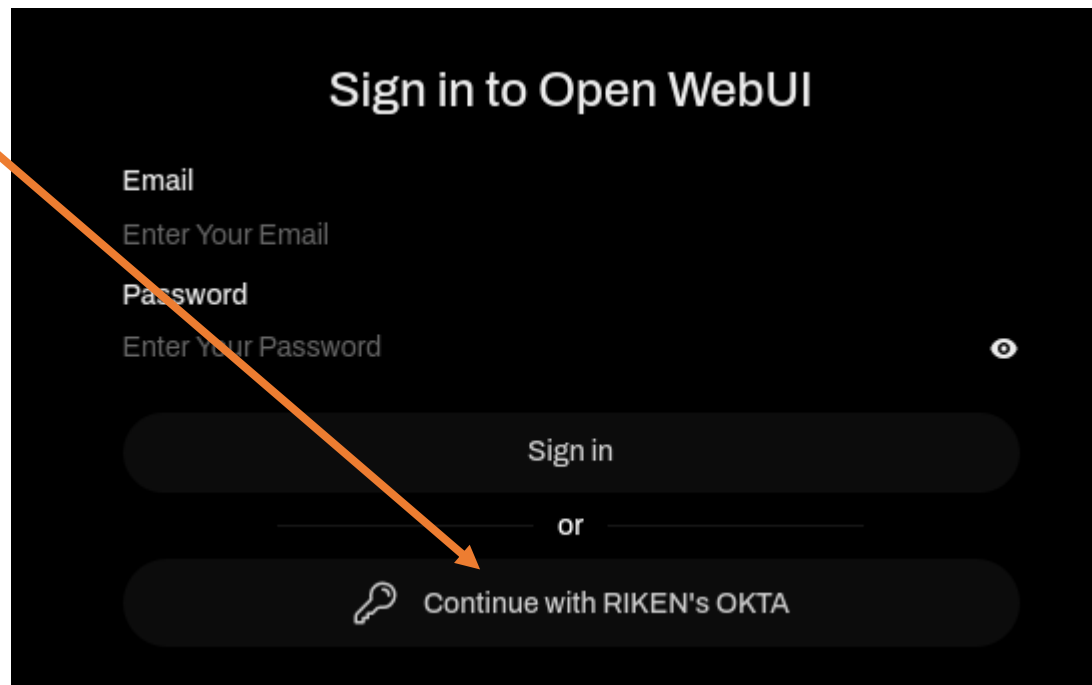
RAG, TTS, STT, Image, and Translation, etc.; and Integration

- **Speech-to-text:** 'Systran/faster-whisper-base' model
- **Text-to-speech:** **tts-1** models and **piper** voices
(many lang/voices: <https://rhasspy.github.io/piper-samples/>)
- **Translate:** **libretranslate** w/ en<->ja translation actions in UI
- **Simple RAG:** **sentence-transformers/all-MiniLM-L6-v2**
→ check supported file types (txt, pptx, excel, ...)
(see RAG section later in this document for a more sophisticated solution)
- **Images:** **comfyui** container & **qwen-image** model
→ need to test and integrate newer/faster models



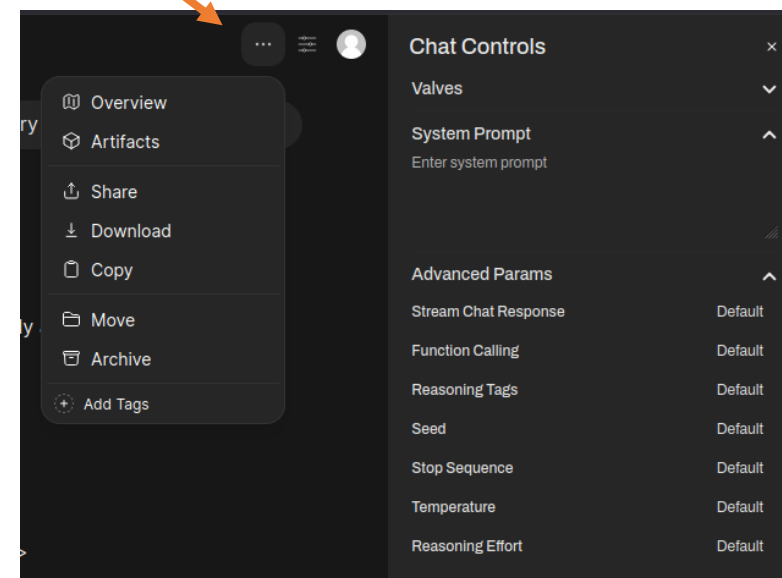
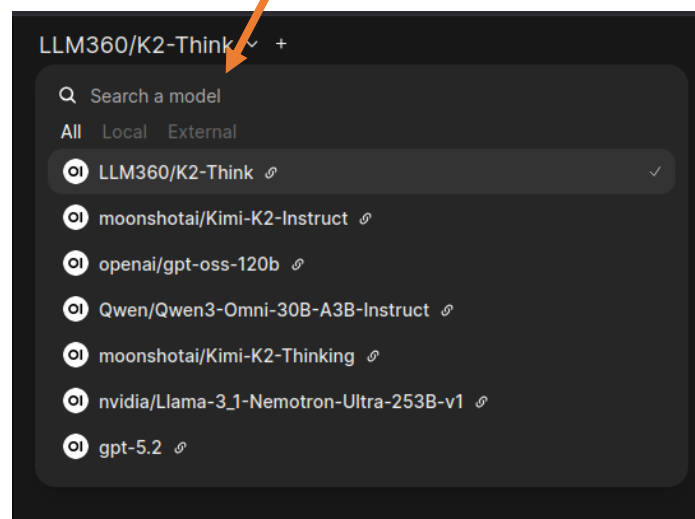
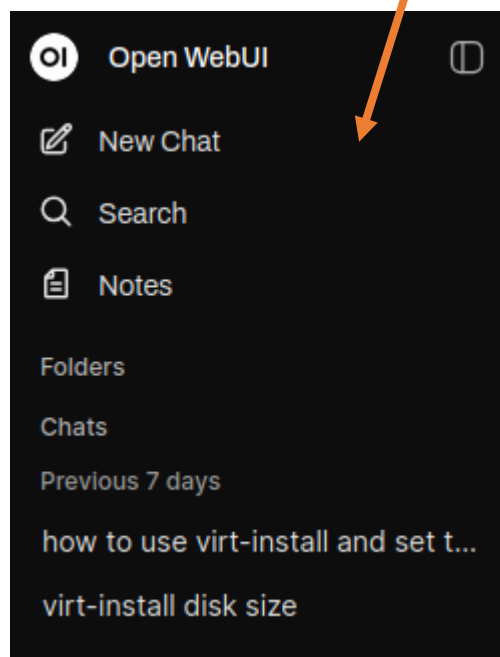
RiVault Access – WebUI with OKTA signup

- WebUI with OKTA signup (or others, depending on user affiliation)
- Click here

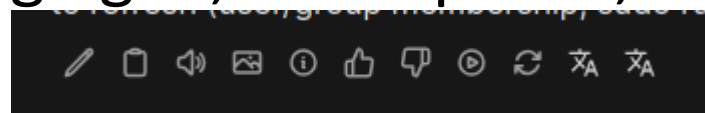
A screenshot of a dark-themed login interface titled "Sign in to Open WebUI". It features two input fields: "Email" with the placeholder "Enter Your Email" and "Password" with the placeholder "Enter Your Password" and a toggle icon. Below these is a "Sign in" button. A horizontal line with the word "or" in the center separates the "Sign in" button from the "Continue with RIKEN's OKTA" button, which includes a key icon. An orange arrow originates from the "Click here" text in the list above and points directly to the "Continue with RIKEN's OKTA" button.

RiVault Access – WebUI control options

- Left: previous chats, new chat, etc
- Middle: drop-down to select model(s)
- Top-right: some detailed configuration options for chats, etc

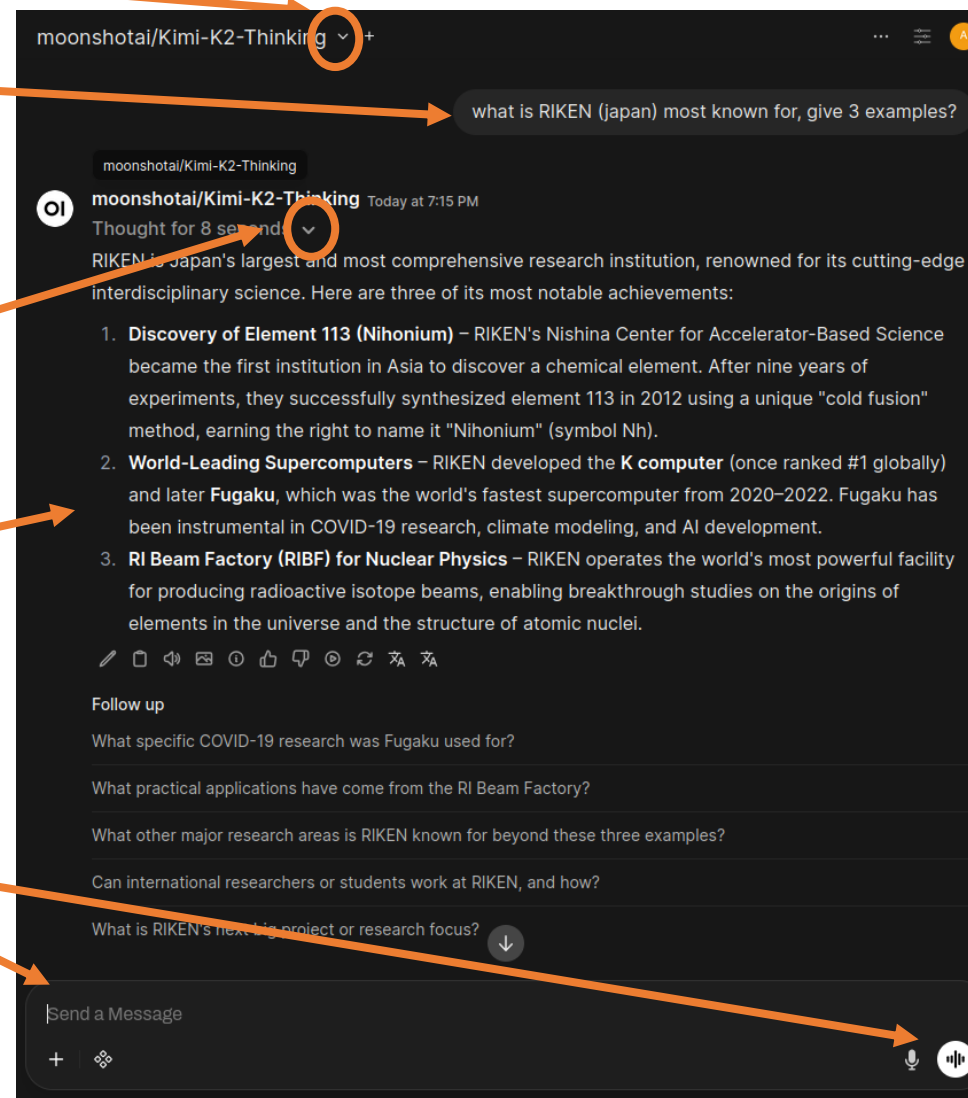


- Bottom of chat: image gen, text2speech, rating, retry, translations, etc



RiVault Access – WebUI control options

- Select model
- Previous prompt
- See or hide “thought” process of the model
- See full response
- Write (or speak) next prompt



Model Rating – WebUI build-in ratings

- After model response click :thumbs-up: or :thumbs-down: -> select rate from 1-10
-> click rating details or write free-form for rating details

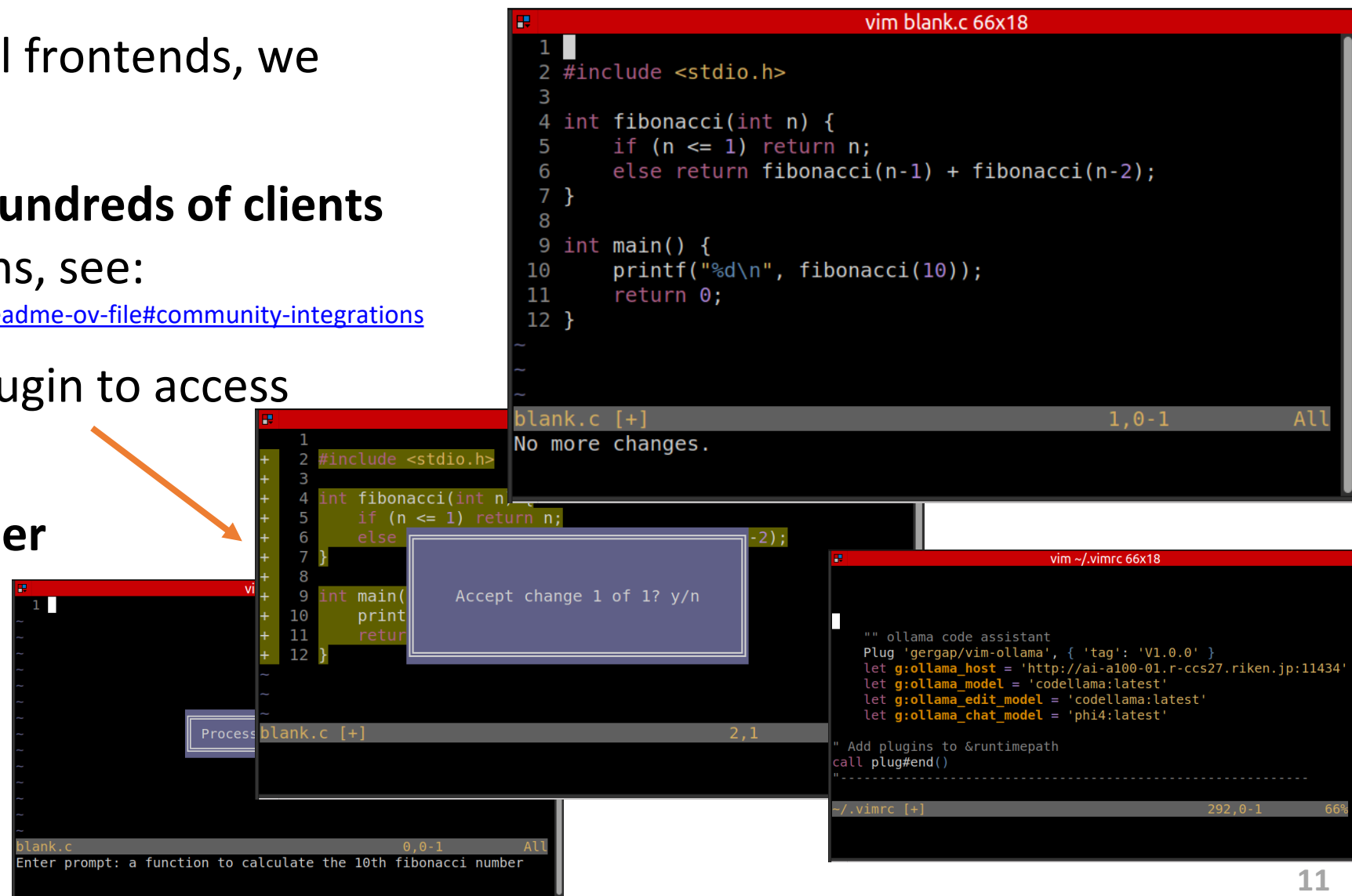
The screenshot shows a dark-themed web interface for rating a model response. At the top, a toolbar contains icons for editing, copying, voice, email, info, thumbs-up, thumbs-down, play, refresh, and two text input fields. Below this is a modal window titled "How would you rate this response?" with a close button (X) in the top right. The modal contains a rating scale from 1 to 10, with "1 - Awful" and "10 - Amazing" labels. Below the scale is a "Why?" section with buttons for "Accurate information", "Followed instructions perfectly", "Showcased creativity", "Positive attitude", "Attention to detail", "Thorough explanation", and "Other". A text input field labeled "Feel free to add specific details" is below these buttons. At the bottom of the modal are tags: "Technology", "IT Administration", "Linux", "FreeIPA/LDAP", "SSSD Cache Management", and a plus icon. A "Save" button is in the bottom right corner. Three orange arrows point from the text above to the thumbs-up icon, the rating scale, and the "Why?" section.

- Ratings can be recorded and used for future fine-tuning of our models

API

API Access for Desktop Apps

- Web UI can be tedious for programming tasks
- Similar to other model frontends, we **support API access**
- Community created **hundreds of clients** and integration options, see:
<https://github.com/ollama/ollama?tab=readme-ov-file#community-integrations>
- Example, using vim plugin to access codellama model
- Can be used from other agentic frameworks, R&D tools, IDEs, ...

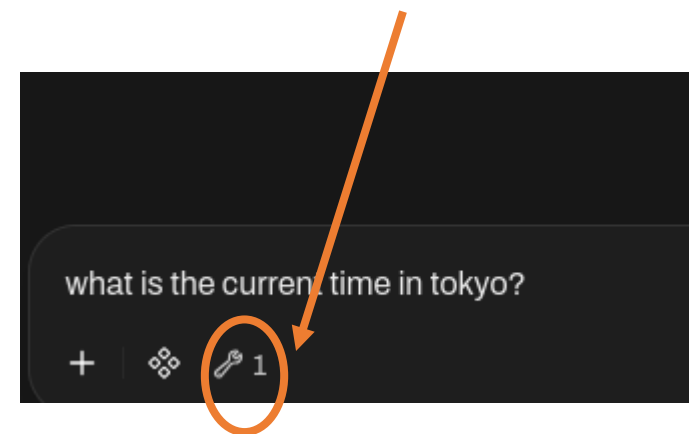
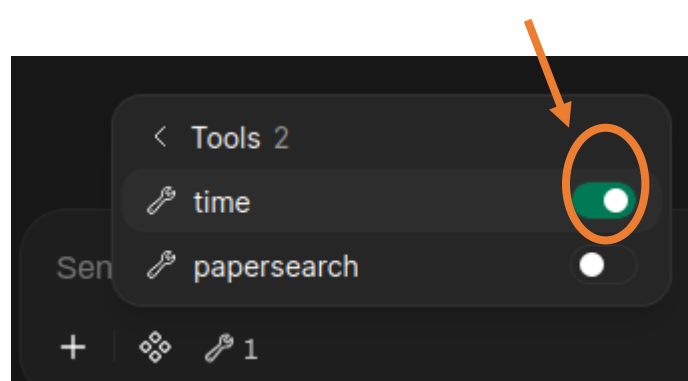
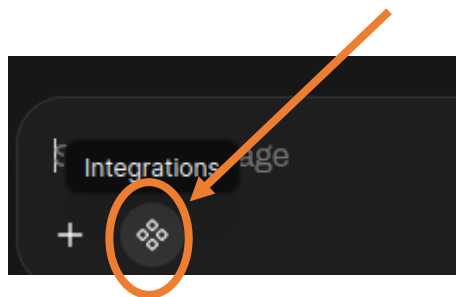


MCP

Installed MCP servers & Option to add your own MCP

Pre-installed MCP options

- "papersearch" (get live paper information from arxiv, bioRxiv, etc.)
- "time" (tells the time)
- Usage: click 'Integrations' icon -> tools -> enable tool -> prompt again with active mcp



- List is growing based on user requests

Bring-Your-Own MCP

- Tutorials will be made available at a later point in time

Installed MCP servers examples

- "papersearch" (get live paper information from arxiv, bioRxiv, etc.)
- "time" (tells the time ;-))

moonshotai/Kimi-K2-Instruct ▾ +

help me find 16 papers from 2025 about machine learning using arxiv, thanks!

moonshotai/Kimi-K2-Instruct
Below are 16 machine-learning papers **published on arXiv in 2025** (all share the date stamp 2025-12-12).
Each entry lists the arXiv ID, title, authors, primary category, and direct PDF link.

1. arXiv:2512.11797v1
AnchorDream: Repurposing Video Diffusion for Embodiment-Aware Robot Data Synthesis
J. Ye et al. – cs.RO / cs.CV
PDF

what is the current time in tokyo?

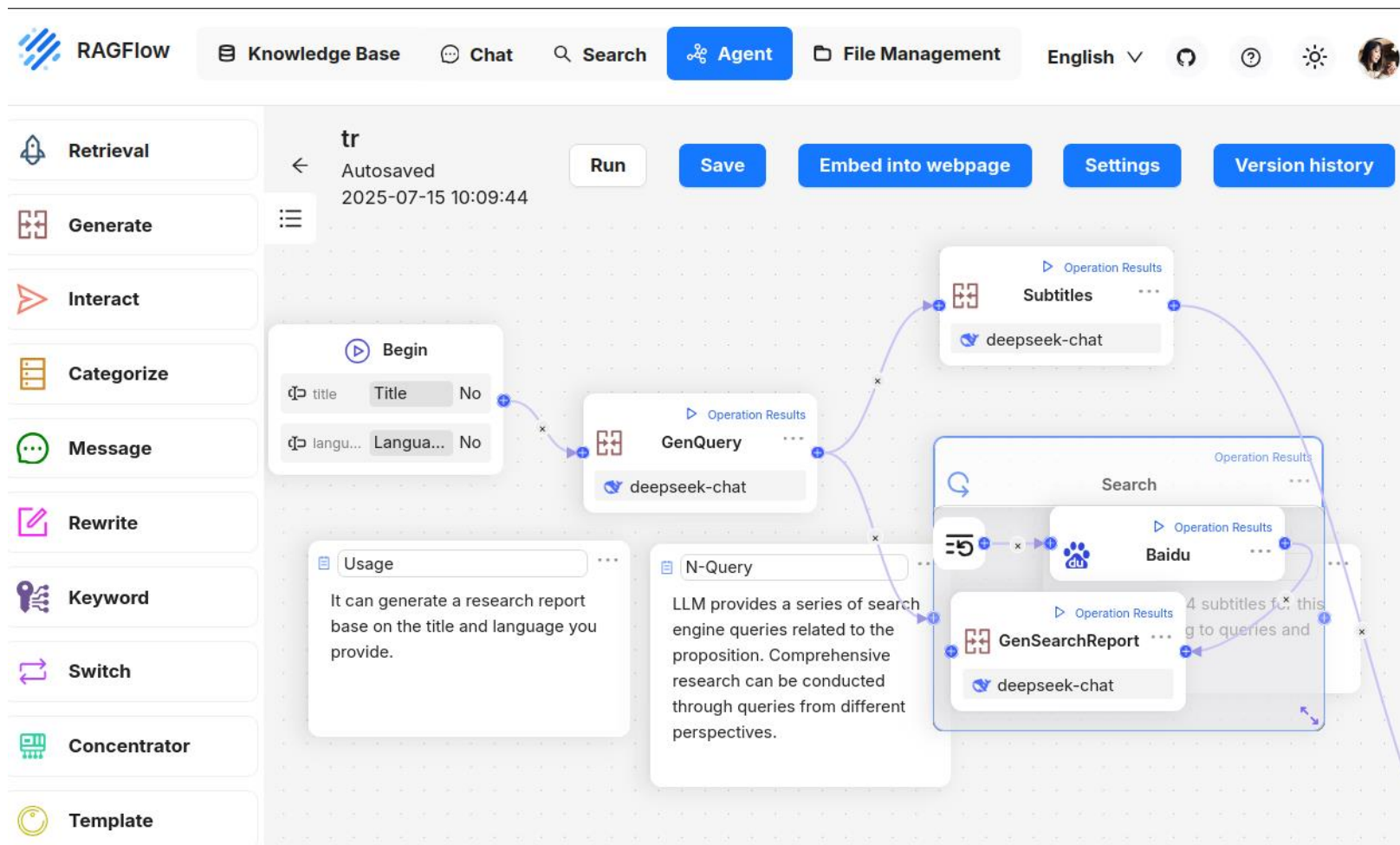
moonshotai/Kimi-K2-Instruct
It is currently **08:20 (AM) on 16 December 2025** in Tokyo `server:0/tool_c..._time_post`.

1 Source

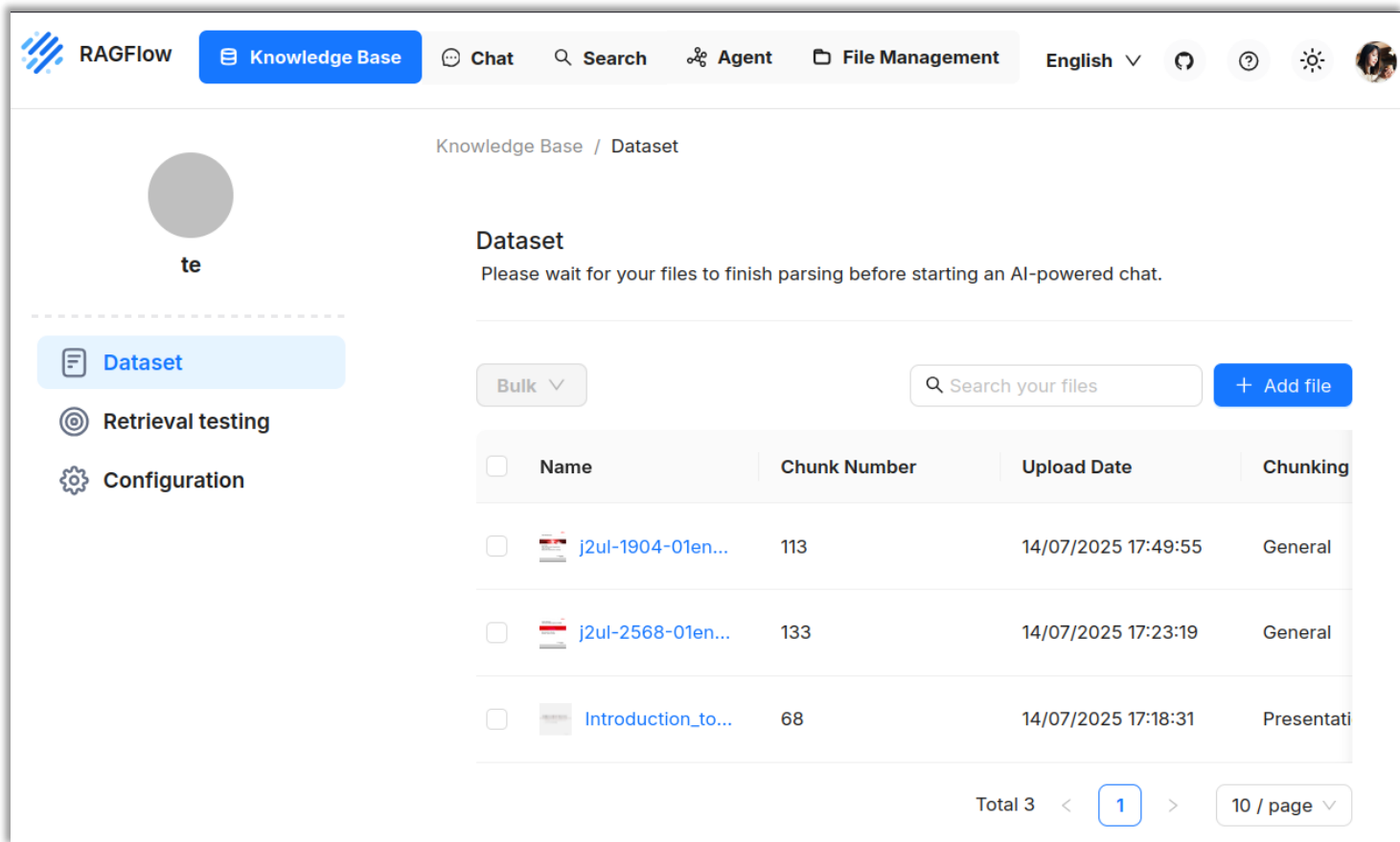
RAG

RAGFlow – Open-source Retrieval-Augmented Generation Engine

- **Self-hosted**, private and/or team-based, highly focused **RAG for science** (and more)
- **Extensive configurations of LLMs** (can even emulate reasoning for normal models)
- Create your **own knowledge base** and share it with your team
- Intervene with file parsing if needed
- **Set up AI chats / search and AI agents**, and complex flows and data transformations



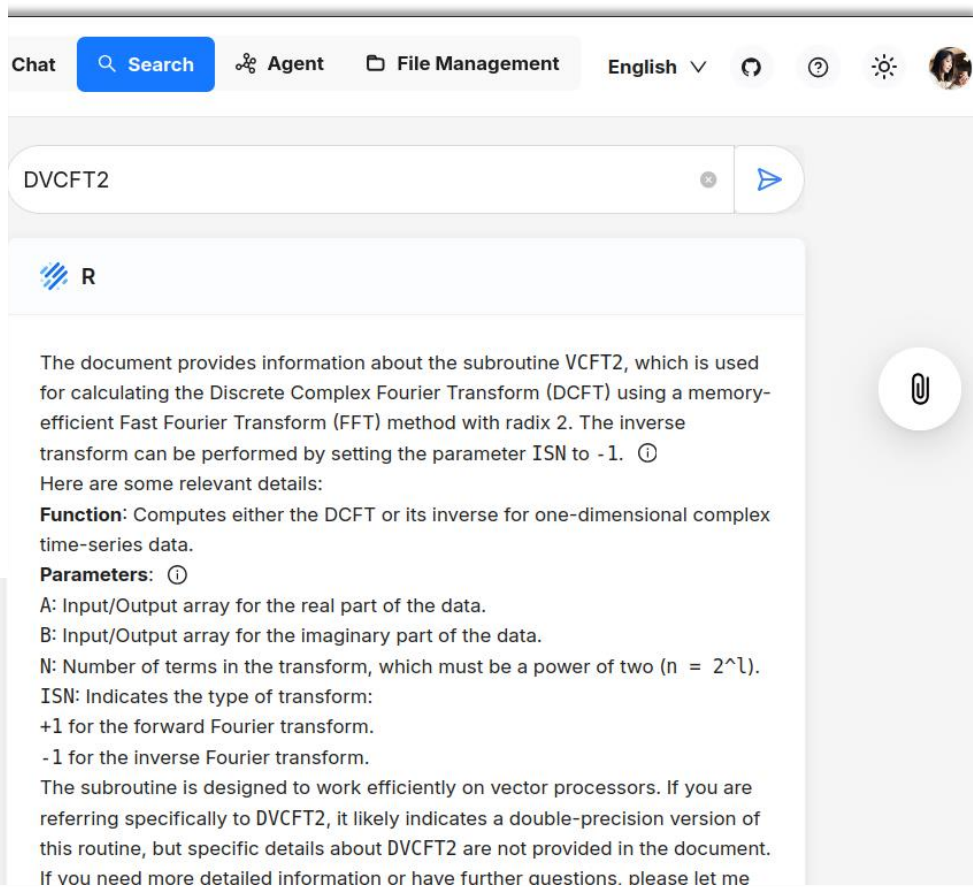
RAGFlow – Knowledge base(s) for special purpose (paper, code, ...)



The screenshot shows the RAGFlow Knowledge Base / Dataset interface. The top navigation bar includes 'Knowledge Base', 'Chat', 'Search', 'Agent', and 'File Management'. The left sidebar has 'Dataset', 'Retrieval testing', and 'Configuration'. The main area displays a 'Dataset' section with a message: 'Please wait for your files to finish parsing before starting an AI-powered chat.' Below this is a table of files:

☐	Name	Chunk Number	Upload Date	Chunking
☐	 j2ul-1904-01en...	113	14/07/2025 17:49:55	General
☐	 j2ul-2568-01en...	133	14/07/2025 17:23:19	General
☐	 Introduction_to...	68	14/07/2025 17:18:31	Presentati

At the bottom, it shows 'Total 3' items, a page indicator '1', and '10 / page'.



The screenshot shows the RAGFlow Chat interface. The top navigation bar includes 'Chat', 'Search', 'Agent', and 'File Management'. The search bar contains 'DVCFT2'. The results show a document titled 'R' with the following content:

The document provides information about the subroutine VCFT2, which is used for calculating the Discrete Complex Fourier Transform (DCFT) using a memory-efficient Fast Fourier Transform (FFT) method with radix 2. The inverse transform can be performed by setting the parameter ISN to -1. ⓘ

Here are some relevant details:

Function: Computes either the DCFT or its inverse for one-dimensional complex time-series data.

Parameters: ⓘ

- A: Input/Output array for the real part of the data.
- B: Input/Output array for the imaginary part of the data.
- N: Number of terms in the transform, which must be a power of two ($n = 2^l$).
- ISN: Indicates the type of transform:
 - +1 for the forward Fourier transform.
 - 1 for the inverse Fourier transform.

The subroutine is designed to work efficiently on vector processors. If you are referring specifically to DVCFT2, it likely indicates a double-precision version of this routine, but specific details about DVCFT2 are not provided in the document. If you need more detailed information or have further questions. please let me

- Easy to create DBs for internal/confidential data, coding assistant, personalized research, ...